

# Minwoo Yoo

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## Education

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### The George Washington University

Sep 2022 – May 2027 (expected)

Ph.D. in Economics

Research Areas: Political Economy, Applied Microeconomics, Text Analysis

### Seoul National University

M.A. in Economics (Empirical IO) — degree requirements completed

Aug 2020

B.A. in Economics, *summa cum laude* (GPA: 3.93/4.0, Top 5%)

Feb 2018

## Working Papers

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### Trade Shocks and Media Slant [Draft]

- Designed a causal inference strategy exploiting regional differences in trade-shock exposure to estimate the effect of economic shocks on local newspapers' political slant.
- Developed article-level measures of political slant by training supervised text classifiers on U.S. Congressional speeches and applying them to 52M newspaper articles.
- Quantified that the share of Republican-leaning articles rose 9.6 percentage points more in most-exposed regions, driven by initially Democratic-leaning papers in low-education areas.

### In and Out: How Do Food Delivery Riders Work? (Draft available upon request)

- Studied how food delivery riders (Uber-like gig workers) adjust participation and hours worked in response to expected wage changes.
- Designed a shift-share IV combining local restaurant composition with national order trends, applied to 21M high-frequency, transaction-level observations.
- Estimated that a 1% wage increase raises hours worked by 2.5–2.8% and the probability of working by 0.7–0.9 percentage points; both responses grow stronger with experience.

## Professional Experience

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### Economic Consultant, Seoul National University

Sep 2018 – Aug 2019

- Prepared analysis submitted to the Korean Fair Trade Commission for the antitrust review of a merger between the world's two largest shipbuilders (Hyundai-Daewoo), with Kim & Chang and Compass Lexecon.
- Built post-merger simulation models of firm-level production and costs; quantified efficiency gains and concluded no expected price increases in Korean government defense contracts.

## Projects

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### Home Credit Default Risk: Imbalanced Classification [GitHub]

Spring 2026

- Built an end-to-end loan default classification pipeline on the Home Credit dataset, covering data preprocessing, feature engineering, model training, and evaluation.
- Compared classical ML models (logistic regression, random forest, gradient boosting) with shallow and deep neural networks, and benchmarked class-imbalance handling techniques such as SMOTE, class weighting, and threshold tuning.

### Political Text Classification [GitHub] [Streamlit Demo]

Fall 2025

- Led an end-to-end machine learning project on political text, covering data preprocessing, feature construction, and model training.
- Built and evaluated two separate models: a TF-IDF + Logistic Regression baseline and a fine-tuned transformer-based model, with hyperparameter tuning and performance comparison.

## Teaching Experience

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### The George Washington University

Sep 2023 – Present

#### Instructor

- Math Camp (Ph.D.), Summers 2024, 2025

Designed and solo-taught the full curriculum (real analysis, linear algebra, optimization, statistics), authoring all lecture notes, problem sets, and exams.

#### Teaching Assistant

- Econometrics I–II (Ph.D.), Fall 2024 – Present

TA for Profs. [Robert F. Phillips](#) and [Ben Williams](#).

- Microeconomic Theory I–II (Ph.D.), Fall 2023 – Spring 2024

TA for Prof. [Sumit Joshi](#).

### Seoul National University

Spring 2018 – Fall 2020

#### Teaching Assistant

- Studies in Industrial Organization (Ph.D.)
- Industrial Organization
- Mathematics for Economics
- Stocks, Bonds, and Financial Derivatives

## Skills

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### Programming & Computing Tools

- Python (pandas, scikit-learn, PyTorch), SQL, R, Stata; AWS, GCP, HPC.

### Causal Inference

- Experimental Design (A/B Testing), Difference-in-Differences (DiD), Synthetic Control, Regression Discontinuity Design (RDD), Instrumental Variables (IV), Propensity Score Matching.

### Structural Econometrics

- Dynamic Programming, Discrete Choice Models.

### Machine Learning

- Logistic Regression, Regularization (Lasso, Ridge), Random Forest, Gradient Boosting, Neural Networks.

### Natural Language Processing

- Transformer-based Models (BERT), Seq2Seq with Attention, RNN (LSTM).

## References

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**Leah Brooks** (Primary Advisor)

Professor of Public Policy and Economics

The George Washington University

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**Alessandra Fenizia**

Assistant Professor of Economics

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